

Uneven Wage Growth and Public Goods

The Case of US Public Education

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Abstract

Wage growth is a key driver of local wealth accumulation, enabling greater household and community investment in public goods. Unfortunately, wage growth has diverged markedly from productivity growth in the United States in recent decades. As such, regions whose wages track productivity gains tend to benefit from broader economic growth, while lagging regions risk weaker savings capacity and declining support for public services. This work estimates the effect of local wage shocks on education spending using a shift-share instrumental variable design, exploiting variation in commuting zone industrial composition interacted with national industry growth shocks. We estimate that a 10% increase in local wages generates a 2.3% short-run increase in per-pupil education expenditure, with a long-run effect of approximately 4.6%. This result is only driven by a handful of states, whereas others exhibit little wage responsiveness. These results expedite the need to enable education equalization programmes to account for future potential wage disparities in key states.

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1 Introduction

Since the 1970s, the United States has experienced a persistent divergence between productivity growth and wage growth [hoffmann2020, stansbury, economicpolicyinstitute, elsby2013]. While labor productivity has continued to rise, the earnings of typical workers have increased far more slowly, leading to a substantial decoupling between the two trends. Explanations range from heterogeneous demand for human capital [autor2014, katz1992, acemoglu2012, autor2011, autor_growth_2013] to competition [stansbury, deb2022, yeh2022, azar2022, wilmers2018] to weakened labor market institutions and policy forces [egger2019, lee1999, mishel2021, barkai2020, autor2016, card2001, bivens2013, stansbury2020].¹ Though its causes are hotly debated, the fact itself is well-documented, especially for lower- and middle-income workers [komlos2018].²

Regardless of its causes, the decline in the wage–productivity link has important consequences. Wages remain the most direct channel through which national productivity growth translates into household well-being. When wage growth lags productivity, improvements in aggregate efficiency do not translate into broad-based improvements in living standards [mishel2021]. Furthermore, in a context in which

¹Summers & Stansbury 2018 argue that productivity growth still exerts a positive influence on wages overall, but that institutional and structural changes have weakened the link for large segments of the workforce [summers2018, stansbury]. They point to declining union density, erosion of the minimum wage, globalization, and increased market concentration as key factors that have shifted bargaining power away from workers and reduced labor’s share of national income [autor2014]. Furthermore, additional evidence finds that this decoupling is far from a universal phenomenon. Rather, decoupling applies almost strictly to lower- and medium-wage earners, while already higher wages manage to keep up (relatively) with productivity growth rates.

²Though authors find that wage inequality growth has stagnated in the last decade, this is not a result of a “catch-up” effect of lower- or middle-income earners with top income earners, but rather wage growth at the bottom of the wage distribution [aeppli2022]. Additionally, further evidence indicates that metric choice can influence the ambiguity of earnings and wage inequality conclusions [parolin2025].

the benefits of economic growth have already accrued unevenly across communities in the United States, this pattern may reinforce existing spatial inequalities in economic well-being, especially in regions where industrial composition and historical shocks have already left communities economically vulnerable [bauluz, rickard2020, fee2025, storper2009, fallah2011, benmelech, greenstone2010].

A link that has been far less explored in this context is the spillover effect of local wages to local wealth-building and its effect on public goods. Wage growth is an important contributor to local wealth-building, allowing households and communities to invest more in local public goods [boustan2013]. Communities whose wages rise in line with productivity growth will likely reap the benefits of economic growth whereas those who do not, risk falling behind. This link is particularly important in the US given the structure of local public financing. Majority of local public services are funded via property taxes. This funding structure entrenches a mechanism for generating inequality of opportunity between diversely affluent regions of the country [manduca2025, weir2021, schecht2024, mettler2011].³ Existing work shows that income inequality and uneven local income growth shape inequalities in local public goods provision, residential mobility, and long-run opportunity [chetty2014]. As such, wage stagnation can translate directly into weakened fiscal capacity. Put plainly, given the structure of US public services, wherein they are tied to asset values, inequality in wealth-building can have significant effects for the quality of local public services.

Community well-being and public expenditure in the US is already characterized by a high degree of spatial heterogeneity [manduca2025, obrien2025, feler2017, chetty2026]. Evidence of how income inequality perpetuates other forms of inequality (opportunity, health, infrastructure quality, and broader well-being) is steadily increasing. [chetty2016, logan2012, semuels2016, Avancena2021, flavin2009, boustanEffectRisingIncome2013] find that greater income inequality leads to higher public expenditure across all public goods indicating that a presence of higher-earners in a local area contributes to higher levels of expenditure. Though this does not support an unambiguous denunciation of inequality in itself, it provides additional evidence for the fact that local incomes affect public expenditure raising the potential for “superstar” and “left behind” regions to emerge absent even income growth.

Local public education is particularly relevant in this context. Public schools are responsible for educating over 80% of American school-age children. In 2019, governments around the US (including the federal government) spent a total of \$870 billion on public education, roughly \$17,013 per pupil [nationalcenterforeducationstatistics2023, skinner2019]. However, the quality of services delivered and fiscal resources spent vary widely across the country [biolsi2022]. Take Connecticut for example. In 2016, according to the Connecticut State Department of Education, the town of Greenwich, one of the highest-income towns in the country, spent \$22,000 per pupil while Bridgeport, although only located 40 kilometers away, spent \$14,000 [semuels2016]. A sometimes lacking, or at best under-performing, federal equalization system perpetuates this inequality structurally [beland2017, biasi2023, hoxby1998, mccabe2019, baker2016, aesa2024].

The quality of public education, especially at an early age, can have long-lasting consequences for personal and economic well-being over an individual’s lifetime as well as generations following them [alfonso2020, zheng2022]. Although only a small piece of the puzzle that determines quality public education, expenditure levels ensure adequate funding for facilities, teachers, administrators, and other services [baker2017, baker2025, jackson2016]. Furthermore, evidence suggests that progressive spending delivers efficiency gains in school performance [rauscher2022]. Therefore, ensuring that local or regional economic decline does not disrupt or weaken education expenditure levels is of paramount importance to ensure greater equality in the long-run. Altogether, this evidence points to the value of identifying the extent to which expenditure on public education is reliant on local wage growth across the country.

Despite the large literature on the existence of wage-productivity decoupling and inequality in local public goods provision, one central link remains under-explored: Are local public education budgets sensitive to local wages, and if so, for whom and under what conditions?

To be more precise, this study investigates the following questions:

³A burgeoning literature points to the role of racial segregation and its enduring legacy in perpetuating inequality in local economic health, wealth, and public service expenditure [avenancio-león2022, trounstine2016, trounstine2021].

RQ1: Do local wage gains affect public education expenditure in levels?

RQ2: If so, is this relationship constant across commuting zones? What sources of heterogeneity mediate this relationship?

RQ3: In light of recent reforms to intergovernmental education funding aimed at reducing disparities in educational expenditure, do intergovernmental transfers mitigate wealth-driven inequalities in public education spending?

1.1 Theoretical Motivation and Empirical Approach

Although as mentioned above, existing work provides descriptive evidence that higher income communities invest more in local services, the causal effect of wage growth on education spending is unclear due to substantial endogeneity. Any study of the linkage between two or more local socioeconomic outcomes presents considerable endogeneity challenges. Higher-wage households select into districts with better schools. Education quality affects local human capital and, thus, wages. Local fiscal decisions respond to political, demographic, and institutional environments that co-evolve with economic conditions. Moreover, regional heterogeneity is large. States differ in their reliance on local versus state funding; commuting zones differ in industrial composition and exposure to national shocks; and localities vary in their ability to translate tax bases into expenditures. On a cultural level, the quality of local education can potentially impact preferences for education in turn. Thus, a national “average” elasticity may be misleading. What is missing is a causally identified, regionally sensitive, and mechanism-informed estimate of the elasticity of public education expenditure to local wages.

To overcome these challenges, we construct a shift-share instrumental variable for local wages. More precisely, to interrogate the elasticity of public education expenditure to wages across US commuting zones, we construct a shift-share instrument that combines fixed local industry employment shares with national industry-level changes in real value added using data from the US Bureau of Labor Statistics and Bureau of Economic Analysis. This instrument generates plausibly exogenous local variation by exploiting how regions are differentially exposed to common national trends, while abstracting from endogenous local dynamics. It is particularly well suited in this setting, since the local tax base, and thus education spending, likely depends on industries that are unevenly distributed across regions but subject to similar industry-specific wage shocks. We use this instrument to identify the effect of wage shocks on local public education expenditure as reported by the US Census Bureau’s Annual Survey of State and Local Government Finances.

Given the substantial heterogeneity across U.S. states arising both from structural sources (such as differences in tax systems, regulatory environments, and legislative institutions) and from evolved characteristics (including industrial composition, income levels, inequality, and broader measures of economic diversity) the scope for identifying a policy-relevant single national average treatment effect is inherently limited. Therefore, we proceed in two steps. First, we provide an initial benchmark using a pooled estimation to establish a baseline relationship between wages and education expenditure that generalizes reasonably across the national economy. Second, we investigate the regional and industrial heterogeneity that these pooled estimates mask via a state-by-state estimation, within-CZ analysis of school district expenditure levels, and grouping commuting zones by their historic wage and GDP growth trajectories to improve comparability of treatment and control groups in our instrumental variable design. In the latter heterogeneity analysis, we construct commuting zone growth rates as idiosyncratic variables absent state and national-level growth rates, understanding that *relative* growth rates are of particular importance in a landscape where local economies determine the actual *value* of wage levels. This allows us to identify where the wage-expenditure link is strongest or weakest. Finally, we evaluate the policy landscape of each state using a database of over 900 state-level legislative and regulatory actions as collated by the Education Commission of the States (ECS) to identify any active policies that might obstruct the transmission channel from wages to public education expenditure. We cross-check the identified redistributive policies against the states in which we do not identify a significant wage-spending relationship despite exhibiting relatively high shift-share instrument variation.

1.2 Main Results and Contribution

1. First, we find that public education expenditure responds strongly and positively to local wage growth. A 10% increase in local wages is associated with a 2.2% increase in public elementary education expenditure, with a long-run increase of 4.6%.
 2. Second, examining potential effect channels, we find that a 10% increase in local wages is associated with a 5% (0.5%) increase in the expenditure of the highest-(lowest-) spending school district in each commuting zone that encompasses more than one school district. This effect is corroborated in a 0.3% increase in the within-CZ variance across school districts as well as a 4.2% increase in the spread between the maximum and minimum spending school district. Essentially, the detected wage effect is mainly attributable to affluent districts pulling away from lower-spending districts, rather than a uniform increase in spending across districts within a single commuting zone.
2. Third, estimating the instrumental variable model separately for each state reveals substantial heterogeneity in the relationship between wages and education expenditure. Only a third of the 40 states analysed in this study show persistent causal non-zero elasticities between wages and education expenditure in a within-state model, indicating that these carry the weight of national-level identification. Across these states the wage elasticity varies between 2-10% increases in response to a 10% increase in local wages, with Colorado, Florida, and South Dakota exhibiting highest wage elasticities. Many of these states rely on an above-average share of education funding coming from local sources rather than intergovernmental transfers, though this is far from a unifying characteristic across selected states. Rather, we find that the presence of a clean fiscal transmission channel from wages to public education expenditure, unobstructed by redistributive policies, defines the statistically significant results in majority of the states that exhibit strong within-state wage-spending relationships. More precisely, comparing states with similarly high relative levels of wage variation across commuting zones, the wage-spending relationship breaks down in the states that employ either property tax recapture and redistribution mechanisms or school district spending caps as tools to achieve greater spending equity.
3. Fourth, elasticities are largest, both in terms of size and statistical significance, in commuting zones with historically low wage growth compared to other regions, and smallest in regions with historically strong wage trajectories. This indicates that regions already experiencing economic stagnation may face compounded disadvantages through weaker fiscal capacity for education spending. Additionally, this evidence points to a delineation between communities who teeter at the margins of inadequate funding versus those whose wages and other economic conditions give stability to local public education systems.

As such, these results make four important contributions to the existing literature. First, the paper provides causal estimates of the elasticity of local public education expenditure to local wage growth across US commuting zones, documenting both the average national relationship and substantial regional heterogeneity.

Second, the results show that local fiscal capacity is closely tied to historically uneven wage growth, implying that wage-productivity decoupling has important implications not only for households but also for public sector institutions that shape long-run opportunity. Third, the paper demonstrates how the structure of the US school financing system amplifies or dampens the effects of local economic change, with weaker equalizing opportunities exacerbating the consequences of wage heterogeneity. In a system in which majority of local public budgets remain reliant on property taxes to fund local public goods, this work demonstrates the importance of explicit redistributive policies from wealthier to less wealthy districts. States such as Texas, New Mexico, Montana, Wyoming, and Arizona, otherwise characterized by substantial wage variation across local commuting zones, insulate school districts from wage-related spending fluctuations through property tax recapture, district- and county-level spending limits, and state equalization guarantees. On the other hand, South Dakota, Colorado, and Florida exhibit comparable wage variation which strongly influences local public education spending. Together, these findings show that the wage-productivity link has implications that go beyond household earnings in certain regions, potentially influenced the inter-generational transmission of economic opportunity in the absence of insulating policies.

In the sections that follow, we outline the various data sources used in Section 2; provide a detailed overview of

our shift-share construction and methodological approach Section 3 with accompanying results; and conclude with a discussion of policy implications in Section 4.

2 Data

We compile a panel dataset of the following indicators across 636 commuting zones (CZ) in 40 US states annually between 2001-2021.

Expenditure and Revenue: This work relies on a harmonized repository from Willamette University of the data collected annually as part of the US Census Bureau’s Annual Survey of State & Local Government Finances (SLGF). The SLGF is the ‘only comprehensive source of information on the finances of local governments in the United States’ [pierson]. The data includes county-level revenue, property taxes, and expenditure on public education including disaggregated values by revenue source (federal, state, or other intergovernmental revenue) and expenditure item (lunches, wages, debt). All values are reported in real US dollars. We aggregate school district measures up to the commuting zone-level to ensure the availability of adequate control and treatment variables. We choose to conduct the analysis on the commuting zone level because it is a more accurate picture of a local labor market area [carpenterWhenUseCommuting2022].⁴ Our main outcome variable is per pupil spending on elementary education.

Population controls: We source commuting-zone level population statistics by aggregating data from county-level populations statistics from the US Census Bureau.

Local GDP: We gather local GDP control variables by aggregating county-level GDP data from the Bureau of Economic Analysis (BEA). This BEA data is only available after 2001, defining the lower limit of our panel’s time dimension. We primarily rely on private industry GDP as a control variable given a large remaining portion of GDP is government expenditure which includes public education expenditure.

Property Prices: The US Federal Housing Finance Agency maintains an annual county-level Housing Price Index (HPI) metric, a geographically linked measure of the movement of single-family house prices. The HPI is a weighted, repeat-sales index, measuring average price changes in repeat sales or refinancings on the same properties. This information is obtained by reviewing repeat mortgage transactions on single-family properties whose mortgages have been purchased or securitized by Fannie Mae or Freddie Mac (two US government-sponsored enterprises that guarantee most US mortgages) [usfederalhousingfinanceagency]. We aggregate to the commuting zone level via a population-weighted mean.

Race Controls: The National Institute of Health National Cancer Institute’s Surveillance, Epidemiology, and End Results (SEER) Program provides annual estimates of total White, Black, American Indian/Alaska Native, Asian/Pacific Islander populations at the county level, optionally by Hispanic or non-Hispanic origin [nationalcancerinstitutedccpssurveillanceresearchprogram]. Though the US Census Bureau provides county-level estimates of racial make-up of local areas in the American Community Survey, this data is unavailable prior to 2009 and is considered less accurate than those provided by the SEER Database.

Adequate Education Spending: The University of Wisconsin’s School Finance Indicators Database provides detailed data on School Funding Adequacy, measured as the dollar estimate needed per pupil to achieve U.S. average test scores [baker].

This data aggregation results in a complete and balanced panel of 636 US commuting zones across 40 states between 2001-2021.⁵ All data used is reported annually at the commuting zone level.⁶ Therefore, apart

⁴The database is provided for six different levels of government: state, county, municipal, township, special district, and school district. Reporting is only mandated in Census years (every five years), and even then missing data remains a challenge. This means that data provided at any other level of government suffers from significant levels of missing data, with a high level of selection bias correlated with administrative capacity. However, strengthened by a partnership with the National Center for Education Statistics, observations for US school districts exhibit near-complete coverage between 1997-2021 (pierson).

⁵20% of US states are missing from the dataset because (1) we impose an exclusion restriction wherein any commuting zone reporting more than five \$0 values for property taxes collected is excluded due to likely measurement error and (2) Connecticut, Maryland, North Carolina, and Virginia have been excluded due to unconventional or incomplete public school district reporting.

⁶In line with similar work on US economic geography, commuting zones were chosen as the unit of analysis as they are a far less arbitrary and more accurate representation of local labor market areas/economies [dorn, fowler2024].

from an indicator of a commuting zone’s state, all variables are time-variant.

Table 1 reports summary statistics across relevant variables. All (dollar) values are reported in (real 2017-chained) thousands except for the House Price Index. Table 2 represents the order of integration of relevant variables calculated using a unit root test designed for heterogeneous panels [im2003]. All variables are integrated I(0) indicating minimal concern for non-stationarity except in the case of enrollment and population.

Table 1: Descriptive Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Enrollment	13,356	62.39	169.90	0.13	3,169.73
Population	13,356	405.18	1,077.99	0.88	18,732.54
Elementary Expenditure per pupil	13,356	11.39	2.99	5.97	58.35
Property Tax per pupil	13,356	3.60	2.43	0.29	32.91
Intergovernmental (IG) Revenue per pupil	13,356	7.12	2.28	1.04	27.50
State IG Revenue per pupil	13,356	6.73	2.03	0.79	26.23
GDP per capita	13,356	44.52	25.27	15.32	388.73
GDP pc - Private Industry	13,356	38.42	25.18	5.85	383.06
House Price Index	12,717	255.18	155.71	85.53	1,947.97
% Black	13,356	0.08	0.12	0.0005	0.72
% White	13,356	0.87	0.14	0.09	1.00
% Hispanic	13,356	0.10	0.15	0.002	0.96

Figure 1 demonstrates the spread of elementary education expenditure per pupil by commuting zone, grouped by state. Each state’s population-weighted mean expenditure per pupil is represented in black. There is considerable within-state variation in per-pupil expenditure levels. Notably, Texas, Montana, and Idaho have commuting zones that spend nearly twice as much as other zones in the same state. Furthermore, the mean level of expenditure is nearly four times as high in the highest-spending state (New York) than in the lowest-spending state, Idaho. We additionally display these values in relation to the estimated “adequate” level of expenditure required for students to achieve average U.S. test scores, represented by the green and red arrows. Majority of states fall short of the deemed adequate expenditure value. Mississippi boasts the greatest shortfall in spending while Wyoming boasts the greatest overshoot in spending. Figure 1 demonstrates the considerable within- and between-state variability in spending rates as well as student needs.

Table 2: Order of Integration

Variable	Order of Integration	I(0) test p-value	I(1) test p-value
Enrollment	I(1)	0.608	<0.0001
Population	I(1)	0.286	<0.0001
Elementary Expenditure per pupil	I(0)	<0.0001	
Property Tax per pupil	I(0)	<0.0001	
Intergovernmental (IG) Revenue per pupil	I(0)	<0.0001	
State IG Revenue per pupil	I(0)	<0.0001	
GDP per capita	I(0)	<0.0001	
GDP pc - Private Industry	I(0)	<0.0001	

Note:

Order of integration determined using the Im-Pesaran-Shin (IPS) panel unit root test with intercept. Lag length selected via AIC with maximum of 4 lags. The null hypothesis is non-stationarity; rejection at the 5% level indicates stationarity. I(0) denotes stationarity in levels and I(1) denotes stationarity in first differences. All variables are log-transformed prior to testing to account for heteroskedasticity.

Education expenditure per pupil by commuting zone

Scattered points: Commuting zone mean expenditure per pupil (2001-2021).

Black diamond: Population-weighted state mean expenditure per pupil (2001-2021).

Triangle: Estimated adequate K-12 expenditure per pupil in 2017. Source: School Finance Indicators Database.

2017-value reported to match the deflator used to convert nominal \$ values to real values in the dataset.

Red (green) indicates whether the state mean is below (above) adequate spending levels.

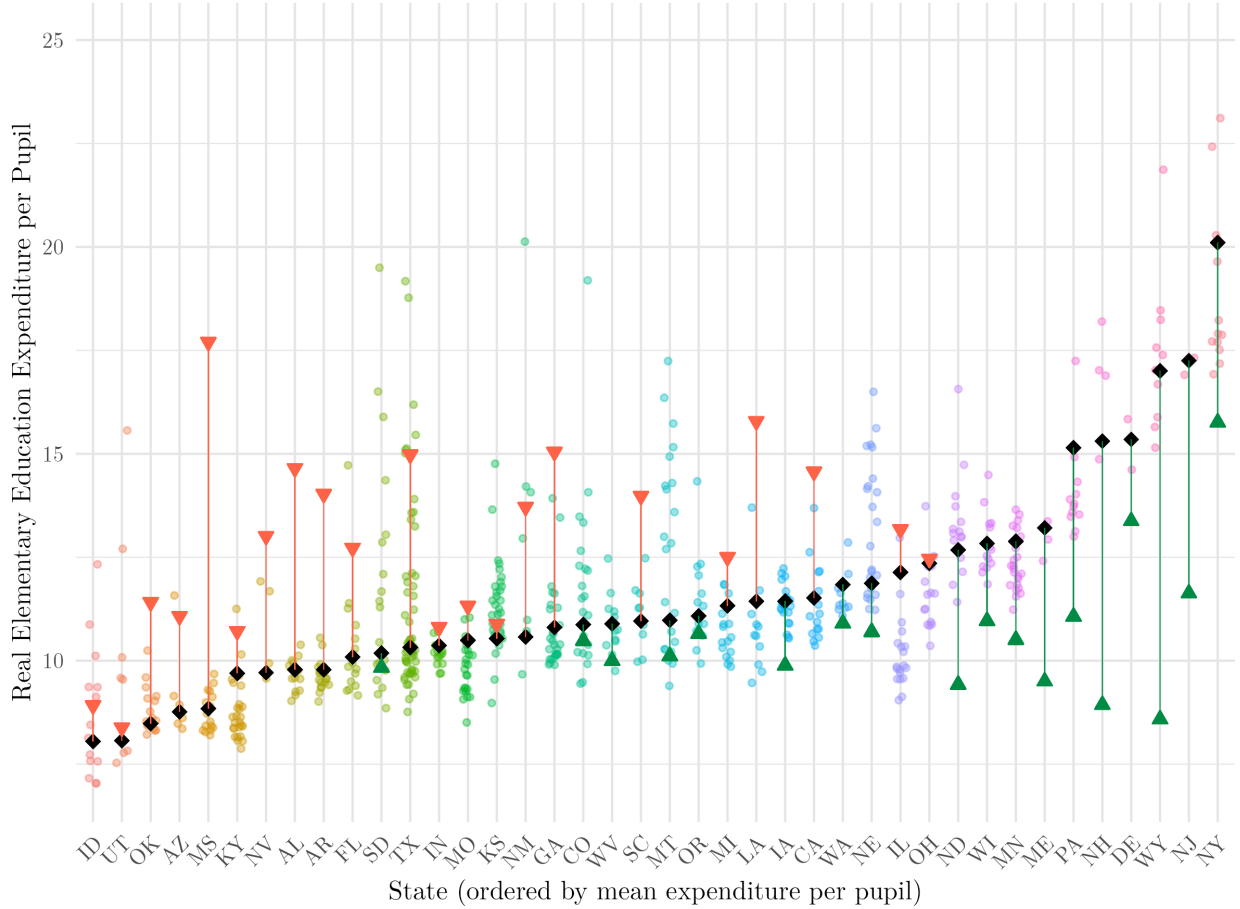


Figure 1: Per Pupil Education Expenditure

Figure 2 demonstrates the trajectories of local wages from an initial level in 2001, demonstrating, the diverging wage growth trajectories of commuting zones in our sample. Blue commuting zones represents those exhibiting relatively high wage growth trajectories as deemed by the calculation outlined in Section 3.2.1. The black dashed line represents the annual national wage growth rate.

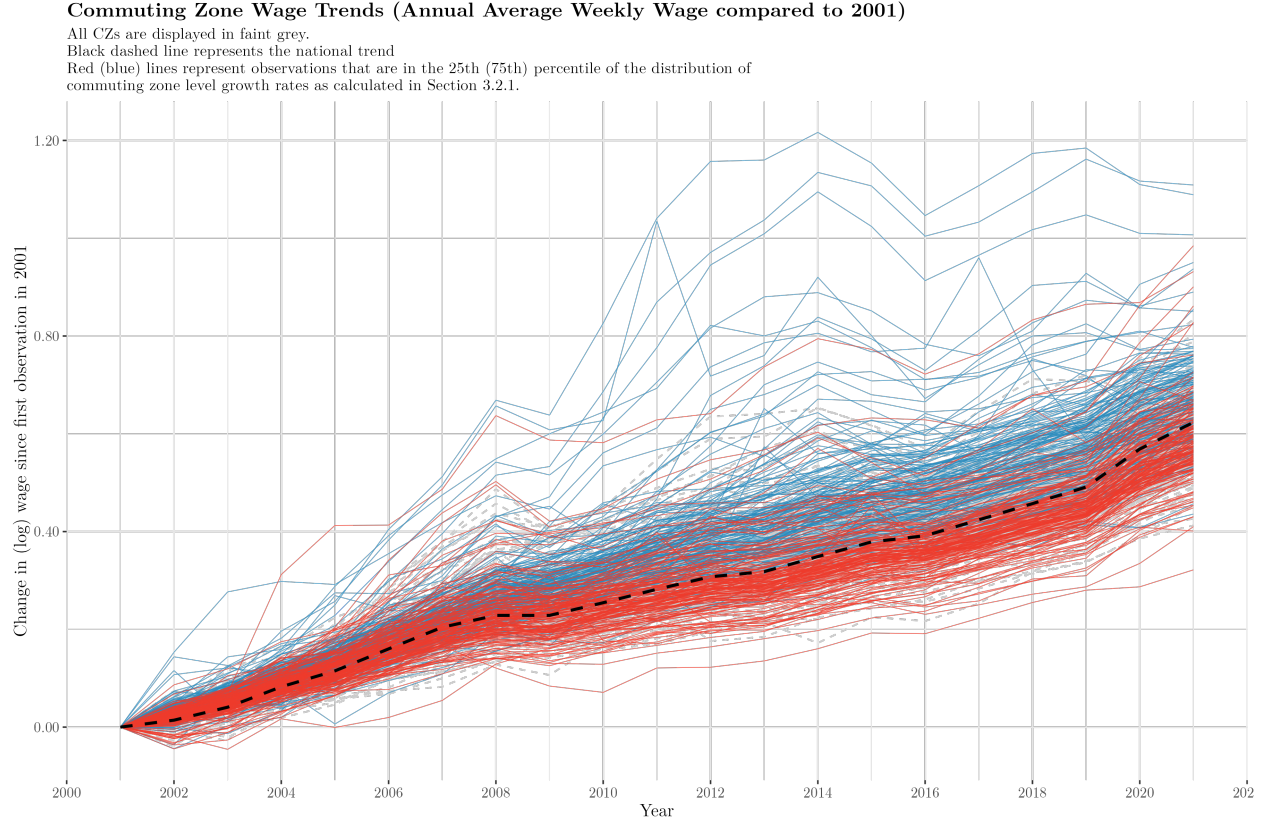


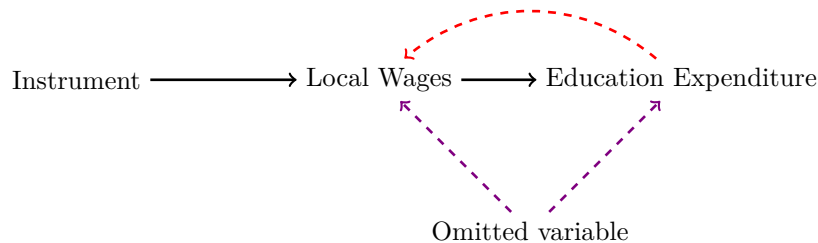
Figure 2: Wages and Productivity

3 Methods

3.1 Identification Strategy

We are centrally interested in the effect of changes in local wages on public education expenditure where there is an evident endogeneity concern between public education expenditure and wages. First, there is a likely attracting factor of high levels of education expenditure for higher-income families. Second, absent migration, education systems provision local labor markets with individuals with diverse human capital.

Figure 3: Instrumental Variable Path Diagram



3.1.1 Shift-share Instrument

Therefore, we adopt a causal identification strategy via a shift-share instrument. Shift-share or *Bartik* instruments have gained popularity in empirical work as a method of handling endogeneity issues in panel data [ferri2022, goldsmith-pinkham2020, bartik1991].⁷ Such instruments combine time-variant, unit-invariant changes in aggregate economic variables (i.e., national changes in industry value added levels) with time-invariant, unit-variant shares in exposure to these macro-level changes (i.e., local shares of employment in particular industries). This decomposition of local-level changes delocalizes variation over space and time. In doing so, it provides a defensible strategy for ‘de-endogenizing’ the treatment, as local exposure is predetermined with respect to contemporaneous shocks. Moreover, by construction, the approach enables the examination of how macro-level phenomena propagate to and affect local units, as it generates local shocks that are driven by national industry trends weighted by the community exposure to these industries.⁸

In the context of this work, we construct the instrument by interacting commuting-zone level industrial employment shares held constant at a base period with national real value added growth by industry. The foundational literature on shift-share instruments outlines various channels through which to isolate plausible exogeneity, each subject to a trade-off between instrument relevance and exogeneity. First, authors argue that local industry shares are exogenous by imposing that shares be fixed to a particular base year and are therefore unable to adapt to changes in national-level growth rates. Such a shift-share instrument is designed as in Equation 1.

$$Z_{it} = \sum_{j=1}^k S_{ij\tau} G_{njt} \quad (1)$$

where $S_{ij\tau}$ is the local share of unit i ’s economy (measured using metrics like employment, wages, revenue) in industry j at a fixed base year τ and G_{njt} is the growth rate of industry j at a national level n at time t .

Alternatively, authors may argue that the claim of exogeneity in the national-level growth rates is unlikely to be violated even when allowing the local shares to vary over time. This approach is likely to come at significant expense to instrument exogeneity. It is constructed as follows:

$$Z_{it} = \sum_{j=1}^k S_{ijt} G_{njt}$$

Finally, authors might be concerned about the implausible exogeneity of both shares and national-level growth rates in which case they construct the instrument as in Equation 2 where the local shares are fixed at a common base year and industry-specific growth rates G are derived from data on other similar regions o rather than national-level changes that are inherently comprised of local-level shifts. This approach likely comes at significant expense to instrument relevance.

$$Z_{it} = \sum_{j=1}^k S_{0jt} G_{ojt} \quad (2)$$

⁷Autor et al. use a shift-share instrument to assess the effect of Chinese import competition on manufacturing employment in US commuting zones [autor2013]. As an extension, [feler2017] use a similar shift-share instrument to assess the effect of the same shock on the size of local government. [baccini2021] employ a shift-share instrument for manufacturing layoffs to tease out the effect of a decline in manufacturing on both economically motivated and racial identity voting patterns in the US.

⁸An additional popular indicator for modelling industrial shocks exogenously is *oil price* as these values are often assumed to be exogenous to local and even national conditions [scheer2022]. Furthermore, various indicators for measuring *deindustrialization* have been proposed including the manufacturing share of employment, value added, and GDP [tregenna2009, tregenna2020]. Finally, in rare instances, exogeneity can be secured due to *geographical, climatological, or geological factors*. For example, [borge2015] obtain an exogenous measure of local revenue by “instrumenting the variation in hydropower revenue, and thus total revenue, by topology, average precipitation and meters of river in steep terrain.” Certain authors have argued that the fact that the location of hydrocarbon deposits is dictated by geomorphological processes provides a plausible argument for exogeneity [esposito2021, chen2022].

Finally, authors can make an additional design choice about whether the effect of these instruments should be assumed common to an aggregate local-level wage growth indicator or allowed to vary by industry. In other words, whether to construct the first-stage relationship of the 2SLS as...:

$$X_{it} = \alpha_i + \beta \sum_{j=1}^k S_{ijt} G_{njt} + \epsilon_{it}$$

...OR...:

$$X_{it} = \alpha_i + \sum_{j=1}^k \beta_j S_j G_{jt} + \epsilon_{it}$$

We employ the formulation in Equation 1, assuming that base-period local industry shares and time-varying national rates are exogenous to local outcomes and construct the former of the first-stage relationships assuming a common β to the sum of these shares.

Using data from the Bureau of Economic Analysis, we construct a shift-share instrument at the commuting zone level using local employment shares by industry and national changes in industry-specific real value added (VA) represented in Equation 3. G_{njt} represents national-level changes in value added in industry j in time t and $\frac{N_{ij\tau}}{N_{i\tau}}$ represents the ‘sensitivity’ of a CZ to these national shocks proxied by an initial share of local employment in industry j in a baseline time period τ . The product of these two values defines the shift-share indicator $\tilde{Z}_{i,t,s}$. In order to construct the share portion, we compute the total local share of employment in a particular industry j . Due to challenges with missing data, we compute an average share across 2001-2005 as our ‘base year’.

We compute the relevant shift-share instrument across 19 two-digit NAICS industrial categories listed in ???. Given industry-level disaggregation of local employment data requires data suppression for anonymity reasons, Figure 2 displays the data coverage of our commuting zone level shift-share instruments. Given the high degree of missingness in the 3-digit categorization we proceed with the 2-digit NAICS codes.

In the Appendices, we provide an additional estimation using a wage-based shift-share instrument constructed using data from the US Bureau of Labor Statistics’ Quarterly Census of Employment and Wages (QCEW). This shift-share instrument is constructed as described above using industry-level changes in real wages. Concerns about endogeneity between the instrument and outcome variable are greater using a wage-based shift-share instrument and is therefore excluded from the main text.⁹

$$\tilde{Z}_{it} = \sum_{j=1}^k G_{njt} \frac{N_{ij\tau}}{N_{i\tau}} \quad (3)$$

3.1.2 Empirical Estimation

This yields a 2SLS AR(1) model defined by the first- and second-stage regressions represented in Equation 4 and Equation 5. Due to the likely presence of time-dynamic effects, we include contemporaneous, 1-year, 2-year time lags of the shift-share instrument.

$$\text{(First stage)} \quad X_{it} = \rho X_{i,t-1} + \phi Y_{i,t-1} + \sum_{\ell=0}^2 \pi_{\ell} \tilde{Z}_{i,t-\ell} + \theta \mathbf{W}'_{it} + \alpha_i^{FS} + \lambda_t^{FS} + u_{it}, \quad (4)$$

$$\text{(Second stage)} \quad Y_{it} = \phi^* Y_{i,t-1} + \beta \hat{X}_{it} + \delta \mathbf{W}'_{it} + \alpha_i^{SS} + \lambda_t^{SS} + \epsilon_{it} \quad (5)$$

⁹We explore the sensitivity of results to the choice of base period τ by constructing the instrument for various base periods as well as a rolling window.

where W_{it} is a vector of control variables. We control for enrollment levels to account for scaling factors in education expenditure, intergovernmental transfers to account for the significant role of such transfers in funding education expenditure, percentage of the population that is Black, percentage of the population that is Hispanic, and private industry GDP per capita levels to account for local price levels.

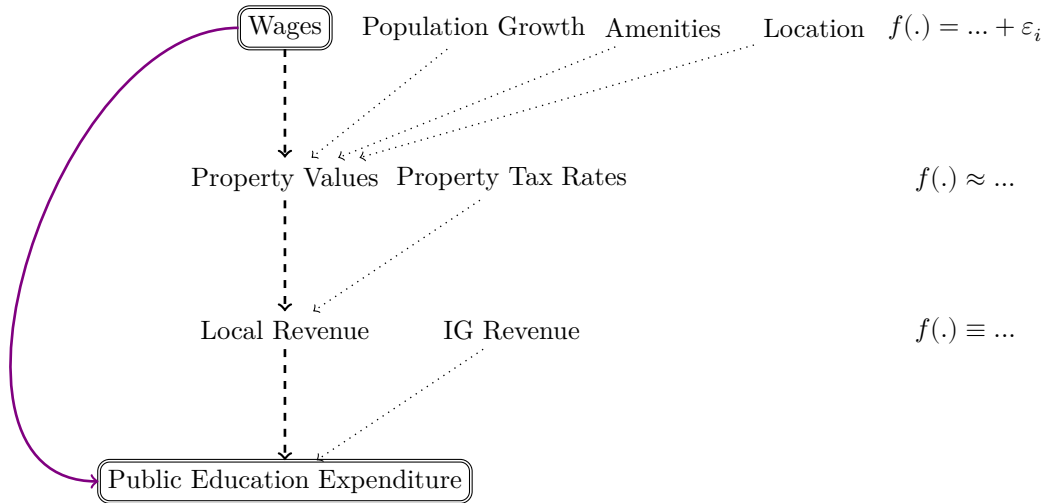
Y_{it} is the natural logarithm of elementary (serving ages 6-12) education expenditure per pupil for CZ i in year t . We focus on elementary education for two reasons. First, this restriction partly shields against a justifiable concern about the endogeneity between wages and quality of local public education. Whereas funding for high school-level education could likely affect local wages given such students are of or near working age, funding for elementary education is unlikely to impact wage rates via a human capital or skills channel. Second, in terms of public impact, elementary education is of foundational importance in the lives of children. Slips in public education provision at a young age could have scarring effects.

α_i represents a CZ fixed effect and λ_t represents year-fixed effects, with stage-relevant superscripts. u_{it} and ε_{it} and represent the error terms of the first and second stage, respectively.

We additionally adopt a dynamic specification by including lagged dependent variables in both stages of the IV estimation to avoid spurious correlation identification arising from persistence in expenditure and wage levels, as well as better accounting for heterogeneity across units. Education spending likely exhibits inertia due to slowly evolving budgetary and relevant policy cycles (i.e., property tax rate setting). Similarly, local wage levels are well-predicted by a previous year's wage levels. Failing to account for these dynamics, our estimates would conflate the causal effect of wage innovations with mechanical persistence in levels. This yields a more conservative and interpretable elasticity, though we demonstrate that the exclusion of the AR(1) term in the second-stage yields a contemporaneous effect estimate nearly identical to the long-run wage effect as derived from the dynamic specification in [Table 3](#).

3.1.3 Interpretation

Figure 4: Channel Diagram



The elasticity of public education expenditure to local wages has an ambiguous interpretation. A positive elasticity would suggest that higher wages increase household savings rates and willingness to invest in local public goods, consistent with standard wealth effects. However, this relationship raises concerns about possible divergence wherein wage growth in high-earning regions could amplify educational investment, potentially widening spatial inequality in public education quality aligning with patterns of income inequality.

Conversely, a negative elasticity could emerge through several channels. Any response in needs-based intergovernmental revenue mechanisms may partially offset local fiscal capacity, creating an inverse relationship between wages and education spending. Alternatively, in more affluent communities, rising wages may enable

households to substitute towards private education, crowding out or reducing demand for public expenditure. Furthermore, such a relationship could provide additional empirical support for a “resource curse” dynamic¹⁰ wherein local communities reprioritize fiscal windfalls toward government expenditure other than public education.

In either case, the consequence of a non-zero elasticity, whether positive or negative, has potential adverse consequences for spatial inequality of public education delivery by either boosting public education in affluent areas or dampening investment in less affluent areas.

Finally, a near-zero elasticity either has a modelling or policy-relevant implication. On the modelling side, a near-zero elasticity could indicate that the wage-public goods relationship operates on a longer time scale than that examined in this work. This would indicate the need for an alternative identification strategy. Alternatively, a near-zero elasticity could indicate that local public education systems are effectively insulated from local wage changes partly because intergovernmental transfers successfully equalize funding across regions.

¹⁰Perhaps the most prominent and often-cited relationship between education and extractive industries is through the lens of the ‘resource curse.’ The validity and empirical existence of a ‘resource curse’ has been tested since its conception with disparate results [ahlerup2020, badeeb2017, blanco2012, borge2015, brunnschweiler2008, cockx2014, cockx2016, deacon2011, dialga2022, douglas2017, haber, menaldo2016, ross2018, ross2015, sincovich2018, wiens2014]. The literature is divided into two strands focusing on either political (the relationship between resource wealth and governance) [ross2015, ross2015, wiens2014, deacon2011] or economic (the relationship between resource wealth and economic growth or human capital) resource curses. Empirical investigation of the economic resource curse has explored the effect of resource dependence on economic growth, public health and education expenditure and outcomes, mainly at a national level [borge2015, cockx2016, douglas2017, sincovich2018]. In the case of education, the distinct outcome measured is level of educational attainment, in other words, whether the presence of a booming resource extraction economy provides disincentives to education for young people. It is worth noting that this literature has been repeatedly questioned on theoretical and conceptual grounds as institutional context often dictates whether a resource curse exists and empirical analyses seem to be very sensitive to methodological choices [ross2018, ross2015, dialga2022]. [borge2015] find support for the paradox of plenty hypothesis in Norway - that higher local public revenue negatively affects the efficiency of local public good provision. [brunnschweiler2008] critically evaluate ‘the empirical basis for the so-called resource curse and find that, despite the topic’s popularity in economics and political science research, this apparent paradox may be a red herring. The most commonly used measure of “resource abundance” can be more usefully interpreted as a proxy for “resource dependence”-endogenous to underlying structural factors. In multiple estimations that combine resource abundance and dependence, institutional, and constitutional variables, we find that (i) resource abundance, constitutions, and institutions determine resource dependence, (ii) resource dependence does not affect growth, and (iii) resource abundance positively affects growth and institutional quality.’ [cockx2014] use a panel on 140 countries from 1995-2009 and find an inverse relationship between resource dependence and public health spending over time. [cockx2016] investigate a panel of 140 countries from 1995-2009 to find an adverse effect of resource dependence on public education expenditures relative to GDP. [dialga2022] find disparate results for health and education controlling for institutional quality. [douglas2017] “measure the effect of resource-sector dependence on long-run income growth using the natural experiment of coal mining in 409 Appalachian counties selected for homogeneity. Using a panel data set (1970–2010), we find a one standard deviation increase in resource dependence is associated with 0.5–1 percentage point long-run and a 0.2 percentage point short-run decline in the annual growth rate of per capita personal income. We also measure the extent to which the resource curse operates through disincentives to education, and find significant effects, but this “education channel” explains less than 15 percent of the apparent curse.’

3.1.4 Results

Data Coverage of Industry-level Employment as Share of Total Reported Employed

Data coverage is calculated as the fraction of total local employment accounted for in the industry-specific employment values. Percentage labels represent the proportion of commuting zones (percentiles) falling below a coverage value using the 2-digit NAICS sectoral classification of industries.

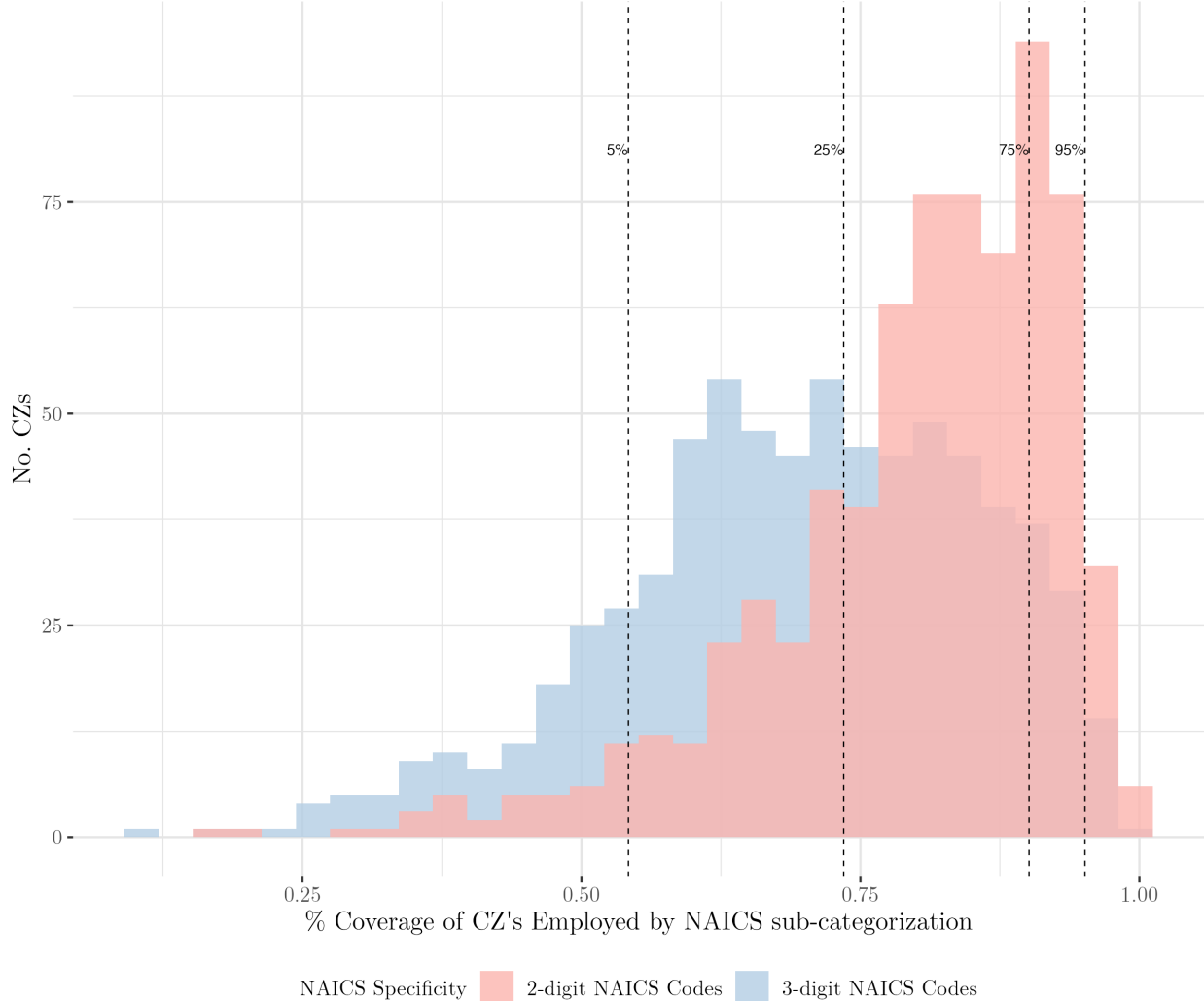


Figure 5: Data Coverage of Industry-level Employment as Share of Total Reported Employed

In [Table 3](#), we demonstrate a strong and highly significant first-stage relationship wherein our shift-share instrument indicates a strong positive contemporaneous relationship with local wages. Our main specification in Columns 1-2 indicates that a 10% increase in local wages leads to a 2.23% increase in per-pupil education expenditure in the short run. The first-stage F-statistic substantially exceeds conventional weak instrument thresholds. The Wu-Hausman test definitively rejects the exogeneity of wages, validating our instrumental variable approach. The implied long-run elasticity is near 0.46, indicating that the cumulative effect of a 10% wage increase is a 4.6% increase in education spending.

In Columns 3-4, we corroborate this long-run effect by removing the AR(1) term in the second-stage regression. The statistically significant causal effect of the treatment approaches this long-run effect of 4.6% (5.6%),

capturing the total association between wages and expenditures, including both the immediate and long-run effects. This near-equivalence in the estimated effect reflects the fact that the underlying data-generating process is defined by dynamics, validating our use of a fully dynamic system.

To contextualize these estimates, we can translate this fiscal response into resource implications. Given a national average per-pupil expenditure of \$18,000 [**nationalcenterforeducationstatistics2023**] in 2021, a 10% wage increase generating a 2-5% spending increase translates to \$350-\$850 per student annually. For a district serving 1,000 students, this \$350,000-\$850,000 increase could support hiring 5-13 additional teachers (assuming an average teacher salary of \$65,000 [**nationalcenterforeducationstatistics2021**]).

Understanding that wage shocks are likely transmitted to education expenditure through property taxes, we test this potential mechanism in columns 5-6 (adjusting the set of control variables to better suit the first-stage relationship in theory). We find a highly significant house price elasticity, wherein a 10% increase in wages generate an 8.4% increase in local house prices. Note that the sample size decreases because of missing data in the housing price index for several of our commuting zones.

Our findings indicate that, in a decentralized education system, local labor market strength affects public education expenditure. Regions experiencing wage growth see spillovers into public education expenditure, whereas communities facing wage stagnation or decline might see their educational spending erode as a result.

Table 3: IV Estimation Using VA-based Shift-share instrument (10, 11, 12) in Levels with CZ and year fixed effects and lags.

Dependent Variables: IV stages Model:	(log) Weekly Wage First (1)	(log) Elem.Ed.Exp.pp Second (2)	(log) Weekly Wage First (3)	(log) Weekly Wage Second (4)
<i>Variables</i>				
(log, 11) Weekly Wage	0.8019*** (0.0099)		0.8038*** (0.0093)	
VA SS (Lvl, 11)	-0.2185*** (0.0483)		-0.2003*** (0.0571)	
VA SS (Lvl, 12)	0.1964*** (0.0445)		0.2105*** (0.0482)	
(11, log) Elem.Ed.Exp.pp	0.0035 (0.0038)	0.5219*** (0.0150)		
(log) IG Revenue pp	0.0125*** (0.0028)	0.2148*** (0.0199)	0.0133*** (0.0026)	
(log) Real GDP Priv. Industry pc	0.0383*** (0.0043)	0.0589*** (0.0143)	0.0385*** (0.0042)	
(log) Enrollment	0.0064* (0.0037)	-0.1729*** (0.0143)	0.0058 (0.0036)	
% Black	-0.1809*** (0.0597)	0.0760 (0.1514)	-0.1800*** (0.0594)	
% Hispanic	0.0045 (0.0250)	0.1265 (0.1198)	0.0041 (0.0249)	
(log) Weekly Wage		0.2088*** (0.0418)		
VA SS (Lvl)			-0.0335 (0.0488)	
<i>Fixed-effects</i>				
unit	Yes	Yes	Yes	
year	Yes	Yes	Yes	
<i>Fit statistics</i>				
Observations	13,356	13,356	13,356	
R ²	0.99328	0.90746	0.99328	
Within R ²	0.79988	0.53452	0.79984	
Wu-Hausman		48.367		
Wu-Hausman, p-value		3.71×10^{-12}		2
Wald (IV only)	2,529.3	24.964	2,244.4	
Wald (IV only), p-value	0×10^{-16}	5.91×10^{-7}	0×10^{-16}	5
F-test (1st stage)	10,125.5		7,963.7	
F-test (1st stage), (log) Weekly Wage		10,125.5		
F-test (1st stage), p-value	0×10^{-16}		0×10^{-16}	
F-test (1st stage), p-value, (log) Weekly Wage		0×10^{-16}		

Clustered (unit) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: IV Estimation Using VA-based shift-share instrument (10, 11, 12) in Levels with CZ and year fixed effects and lags.

Dependent Variables: IV stages Model:	(log) Weekly Wage First (1)	(log) House Price Index Second (2)	(log) Weekly Wage First (3)	(log) Prop Tax pp Second (4)	(log) Weekly Wage First (5)	% Enrolled in Private School (ACS5) Second (6)	(log) Weekly Wage First (7)	% Enrolled in Private School (ACS1) Second (8)
<i>Variables</i>								
(log, 11) Weekly Wage	0.7938*** (0.0103)		0.8020*** (0.0109)		0.6670*** (0.0267)		0.7099*** (0.0216)	
VA SS (Lvl, 11)	-0.1871*** (0.0423)		-0.2299*** (0.0545)		-0.3578*** (0.1081)		0.1084* (0.0616)	
VA SS (Lvl, 12)	0.1824*** (0.0396)		0.2124*** (0.0502)		0.2274** (0.1014)		-0.1220* (0.0641)	
(log, 11) House Price Index	0.0132*** (0.0040)	0.8268*** (0.0088)						
(log) Real GDP Priv. Industry pc	0.0409*** (0.0059)	0.0318*** (0.0077)	0.0394*** (0.0043)	0.0607*** (0.0177)	0.0404*** (0.0063)	-0.0096** (0.0045)	0.0683*** (0.0078)	0.0015 (0.0112)
% Black	-0.1159* (0.0595)	-0.5002*** (0.1306)	-0.2140*** (0.0645)	0.3914 (0.2699)	-0.4214** (0.1852)	-0.1009 (0.1046)	-0.0847 (0.0779)	-0.2550 (0.2268)
% Hispanic	0.0050 (0.0262)	0.0440 (0.0554)	-0.0038 (0.0280)	-0.4805** (0.2044)	-0.0252 (0.0805)	0.0647 (0.0560)	-0.1357** (0.0527)	0.0610 (0.1060)
(log) Weekly Wage		0.1324*** (0.0365)		0.2658*** (0.0487)		0.0377* (0.0192)		0.0590 (0.0386)
(log) Prop Tax pp (11)			-0.0039 (0.0025)	0.7177*** (0.0155)				
(log) IG Revenue pp			0.0124*** (0.0027)	-0.1193*** (0.0171)	0.0153*** (0.0045)	-0.0081** (0.0034)	0.0221*** (0.0040)	-0.0068 (0.0065)
(log) Enrollment			0.0060 (0.0041)	-0.2723*** (0.0226)	0.0251*** (0.0076)	-0.0129* (0.0072)	0.0376*** (0.0064)	-0.0225** (0.0106)
(11) % Enrolled in Private School (ACS5)					-0.0317 (0.0263)	0.6763*** (0.0244)		
(11) % Enrolled in Private School (ACS1)							-0.0067 (0.0078)	0.0301 (0.0247)
<i>Fixed-effects</i>								
unit	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>								
Observations	12,689	12,689	12,720	12,720	6,360	6,360	4,035	4,035
R ²	0.99402	0.99032	0.99288	0.97611	0.98973	0.91947	0.99571	0.68561
Within R ²	0.80863	0.79431	0.79636	0.64944	0.53062	0.44733	0.73519	0.00535
Wu-Hausman		75.484		25.197		19.939		0.12254
Wu-Hausman, p-value		4.37×10^{-18}		5.25×10^{-7}		8.15×10^{-6}		0.72631
Wald (IV only)	2,002.9	13.143	2,233.0	29.757	387.61	3,8434	374.21	2.3356
Wald (IV only), p-value	0×10^{-16}	0.00029	0×10^{-16}	4.99×10^{-8}	3.6×10^{-211}	0.04999	2.7×10^{-214}	0.12653
F-test (1st stage)	8,442.4		9,289.6		1,704.2		1,996.8	
F-test (1st stage), (log) Weekly Wage		8,442.4		9,289.6		1,704.2		1,996.8
F-test (1st stage), p-value	0×10^{-16}		0×10^{-16}		0×10^{-16}		0×10^{-16}	
F-test (1st stage), (log) Weekly Wage		0×10^{-16}		0×10^{-16}		0×10^{-16}		0×10^{-16}
F-test (1st stage), p-value, (log) Weekly Wage								
<i>Clustered (unit) standard-errors in parentheses</i>								
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>								

Table 5: Within-CZ Outcomes via IV Estimation Using VA-based shift-share instrument.

Dependent Variables: IV stages Model:	(log) Weekly Wage First (1)	Variance Expenditure pp Second (2)	(log) Weekly Wage First (3)	Minimum District Expenditure pp Second (4)	(log) Weekly Wage First (5)	Maximum District Expenditure pp Second (6)	(log) Weekly Wage First (7)	Max-Min District Expenditure pp Second (8)
<i>Variables</i>								
(log, 11) Weekly Wage	0.8003*** (0.0107)		0.8000*** (0.0107)		0.8009*** (0.0107)		0.8009*** (0.0107)	
VA SS (Lvl, 11)	-0.2275*** (0.0553)		-0.2299*** (0.0553)		-0.2287*** (0.0551)		-0.2293*** (0.0552)	
VA SS (Lvl, 12)	0.2016*** (0.0512)		0.2009*** (0.0511)		0.2037*** (0.0508)		0.2039*** (0.0509)	
Variance Expenditure pp (11)	-0.0008 (0.0005)	-0.0099 (0.0142)						
(log) IG Revenue pp	0.0143*** (0.0028)	0.0096 (0.0475)	0.0142*** (0.0028)	0.2235*** (0.0421)	0.0145*** (0.0028)	0.1998*** (0.0372)	0.0145*** (0.0028)	0.2694*** (0.0743)
(log) Real GDP Priv. Industry pc	0.0387*** (0.0044)	-0.0037 (0.0471)	0.0387*** (0.0044)	0.0737* (0.0418)	0.0388*** (0.0044)	0.0978*** (0.0301)	0.0388*** (0.0044)	0.1224*** (0.0592)
(log) Enrollment	0.0090** (0.0040)	0.2888** (0.1211)	0.0091** (0.0040)	-0.5963*** (0.1052)	0.0087** (0.0040)	-0.1697*** (0.0463)	0.0087** (0.0040)	-0.1206 (0.0953)
% Black	-0.2013*** (0.0654)	-1.209** (0.5501)	-0.2011*** (0.0654)	1.700* (0.8673)	-0.2019*** (0.0654)	-0.1285 (0.6443)	-0.2032*** (0.0653)	-0.8268 (1.365)
% Hispanic	0.0019 (0.0293)	-0.2800 (0.5122)	0.0018 (0.0293)	0.6480 (0.6286)	0.0028 (0.0292)	0.1597 (0.4101)	0.0027 (0.0291)	-0.1929 (0.7361)
(log) Weekly Wage		-0.2816* (0.1485)		0.9593*** (0.1921)		0.2241** (0.0953)		0.3242* (0.1892)
Minimum District Expenditure pp (11)			0.0005 (0.0003)	0.0473** (0.0216)				
Maximum District Expenditure pp (11)					-0.0012 (0.0008)	0.5066*** (0.0249)		
Max-Min District Expenditure pp (11)							-0.0011** (0.0004)	0.4282*** (0.0181)
<i>Fixed-effects</i>								
unit	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>								
Observations	12,540	12,540	12,540	12,540	12,540	12,540	12,540	12,540
R ²	0.99291	0.13959	0.99291	0.32790	0.99291	0.81654	0.99291	0.79719
Within R ²	0.79171	0.06510	0.79170	0.02965	0.79174	0.28145	0.79185	0.19369
Wu-Hausman		0.37559		6.0907		10.408		6.8160
Wu-Hausman, p-value		0.53999		0.01360		0.00126		0.00905
Wald (IV only)	2,305.5	3.5980	2,312.6	24.933	2,317.3	5.5313	2,324.6	2.9360
Wald (IV only), p-value	0×10^{-16}	0.05787	0×10^{-16}	6.01×10^{-7}	0×10^{-16}	0.01869	0×10^{-16}	0.08665
F-test (1st stage)	9,419.8		9,359.9		9,422.0		9,440.8	
F-test (1st stage), (log) Weekly Wage				9,359.9		9,422.0		9,440.8
F-test (1st stage), p-value	0×10^{-16}		0×10^{-16}		0×10^{-16}		0×10^{-16}	
F-test (1st stage), p-value, (log) Weekly Wage		0×10^{-16}		0×10^{-16}		0×10^{-16}		0×10^{-16}
<i>Clustered (unit) standard-errors in parentheses</i>								
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>								

3.2 Accounting for Heterogeneity

In order to make meaningful policy-relevant insights, we need to unmask the substantial heterogeneity obscured by national-level average treatment effects. These national-level estimates are unlikely to apply uniformly across states and commuting zones, especially given heterogeneity in local tax regimes.

Therefore, we (1) use data on local wages and GDP to create indicators for whether regions have exhibited relative decline or growth compared to other commuting zones to partition our sample in a data-driven manner, employ (2) state-by-state and (3) industry-by-industry estimations in our IV specifications using our VA-based shift-share instrument.

For completeness, we provide results of average treatment effects for all implemented estimations in the Appendices.

3.2.1 Declining vs. Growing Regions

First, we identify declining and growing regions by estimating commuting-zone wage and private industry GDP growth rates conditional on state and national level growth rates and partition our sample across this distribution.

In order to identify declining and growing commuting zones, we estimate separate time series models by commuting zone as follows. These models allow for the identification of commuting-zone level growth rates while controlling for state and national trends in a two-step framework. First, we orthogonalize the state-level growth rate with respect to the national trend, isolating state-specific fluctuations unrelated to the national business cycle:

$$\Delta \log \widetilde{GDPpc}_t^{state} = \Delta \log GDPpc_t^{state} - \hat{\gamma} \Delta \log GDPpc_t^{nat}$$

Second, we regress commuting zone growth on both the national growth rate and the orthogonalized state residuals, thereby decomposing local growth into national, state, and idiosyncratic components. This approach identifies commuting zones whose trajectories systematically diverge from higher-level aggregate patterns, providing a clean measure of relative local economic performance.

$$\Delta \log GDPpc_t^{CZ} = \alpha_g + \beta_n \Delta \log GDPpc_t^{nat} + \beta_s \Delta \log \widetilde{GDPpc}_t^{state} + \varepsilon_t$$

In these equations, each GDP term represents the private industry GDP per capita at the CZ, state, or national level, denoted by superscript.

Intuitively, this specification measures how much of each CZ's growth can be explained by broader aggregate trends versus localized factors. By controlling for orthogonalized state and national variation, the estimated intercept (α_g) and residual terms capture persistent, region-specific trends that are not driven by common macroeconomic forces. This allows us to identify which commuting zones are systematically growing or declining relative to their state and national baselines, thereby providing a purer measure of local economic dynamics that is robust to shared higher-level shocks.¹¹

We perform the same trend deviation calculation for wages where each wage variable represents the commuting zone, state, and national level growth rate in the weekly average wage as reported in QCEW.

$$\Delta \log \widetilde{Wage}_t^{state} = \Delta \log Wage_t^{state} - \hat{\gamma} \Delta \log Wage_t^{nat}$$

$$\Delta \log Wage_t^{CZ} = \alpha_w + \beta_n \Delta \log Wage_t^{nat} + \beta_s \Delta \log \widetilde{Wage}_t^{state} + \varepsilon_t$$

Figure 5 plots the distribution of values of α_g , α_w , and the distribution of commuting-zone level loadings on national and state-level growth rates. The figure demonstrates that commuting zones load more variably

¹¹We provide similar analysis of gross GDP in the Appendix.

onto state-level growth rates and more consistently onto national-level growth rates. These distributions are expected by design as the state-level growth rates provide variation that the national level rates do not.

Next, Figure 6 demonstrates the considerable variability in GDP-level growth rates across commuting zones in the US between 2001-2021. Visualizing the per capita growth rate deviations by state and region demonstrates heterogeneity in this variability across states and regions. For example, Texas, Montana, North Dakota, and Colorado have outstanding positive outliers in the distribution whereas Kentucky, Louisiana, South Dakota have outstanding negative outliers. The grey lines represent the commuting zones value of α_w , indicating that in a large share of cases, wage growth rates are defined by a different sign than the GDP per capita growth rates, indicating local divergence between these rates. Though the calculations above could lead to insignificant such relationships because the growth rates are calculated in reference to different data (indeed the distributions represented in the top left panel of Figure 5 have different median values), the volume of states exhibiting diverging wage and GDP per capita growth rates indicate that such a divergence is likely a fact of life in many commuting zones.

To account for this inherent incomparability of the growth rates α_w and α_g , we calculate a standard Pearson correlation coefficient between the commuting zone time series of GDP per capita and wages, indicating that several states house commuting zones whose wages do not track GDP growth. This analysis is presented in the appendix.

Commuting Zone Growth Rates and Jurisdictional Loadings

Each point or component unit of a distribution represents a single commuting zone.

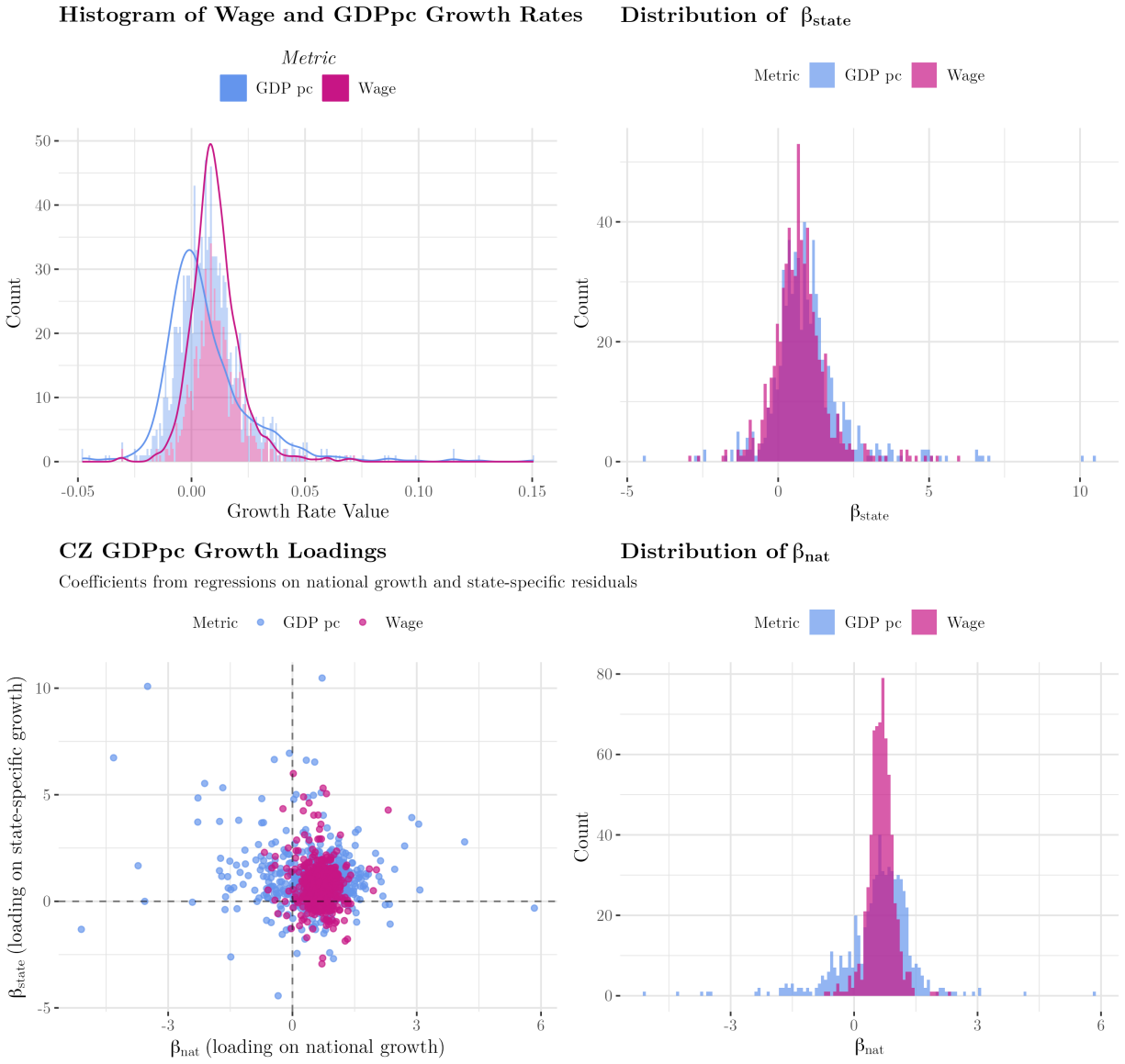


Figure 6: GDPpc and Wage Growth Rates and Loadings

Commuting Zone GDP pc and Wage Growth Rates

Commuting zone-level wage and GDP per capita growth rates are calculated using annual data from 2001-2021, controlling for national and state-level residual growth rates. The GDP per capita trends are represented in order by the colored bars/blue dots, grouped either by region (left), state (center), or overall (right). The grey bars represent the same commuting zone's wage growth rate.

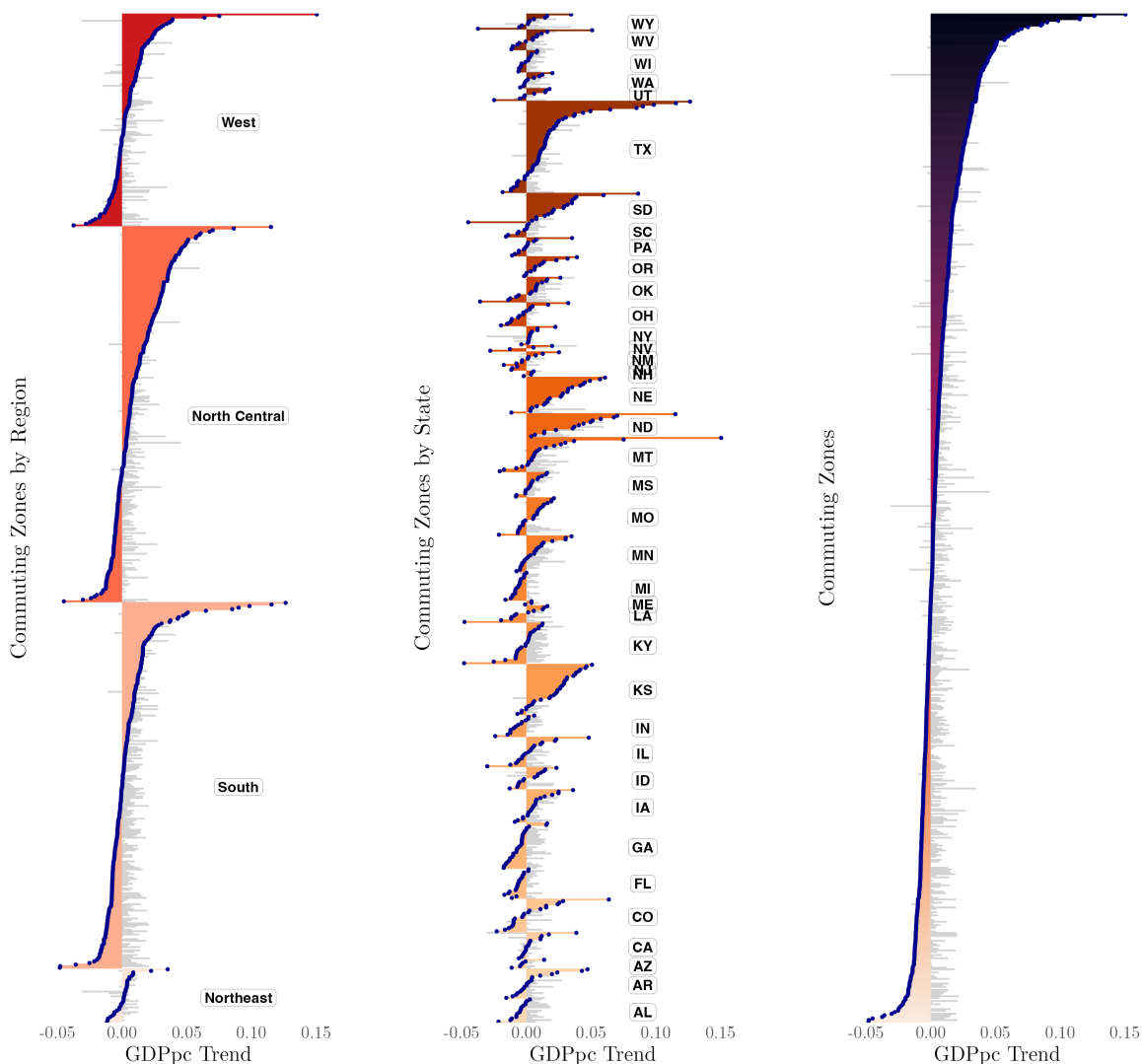


Figure 7: Lollipop Plot of Wage and GDPpc Growth Rates

3.2.1.1 Sample Partitioning by Growth Rates

Using these growth rates, we partition the sample according to the percentiles described above. [Table 7](#) and [Table 8](#) examine how the relationship between local economic conditions and elementary education expenditure per pupil varies across structurally growing and declining regions as defined in the previous section. We partition our sample into four sub-samples by their values of α_w and α_g as shown in [Table 6](#), where the “Declining” and “Growing” sub-groups are inclusive of the “Hyper-Declining” and “Hyper-Growing” communities, respectively.

Table 6: Category Definitions

Category	Definition (α_w, α_g)
Declining	$\alpha < 0$
Hyper-Declining	$\alpha < P_{25}$
Growing	$\alpha > 0$
Hyper-Growing	$\alpha > P_{75}$

Zones with negative (positive) values of α_w or α_g are designated as declining (growing), while those in the bottom (P25) and top (P75) quartiles are labelled hyper-declining and hyper-growing, respectively. This stratification enables comparison of fiscal responsiveness across local economies with different long-run growth trajectories.

Table 7 partitions CZs by α_w . Interestingly, we observe positive statistically significant relationships between wages and public education expenditure though the magnitude and statistical significance of this relationship declines almost uniformly as α_w discretely increases. We see a similar declining size and significance in the explanatory variable accounting for inter-governmental revenue, indicating that these revenues play a smaller role in regions exhibiting above-average wage growth.

Table 8 partitions CZs by long-run GDP per capita trends. The statistical significance and magnitude variations in the wage response tell a less monotonic story here, likely mirroring the lack of systematic correlation between α_w and α_g .

Regardless, all models except Column 5 in Table 7 remain well-identified, with high first stage F statistics, convincing performance on the Wu-Hausman endogeneity tests and Wald tests. Combined, these results indicate that partitioning by background wage growth rates provides greater insight into the sample's heterogeneity than diversity in GDP per capita growth rates. More precisely, given communities feel wage growth more directly than GDP per capita improvements, it is likely that the relationship between public education and wage changes is better captured by sub-sampling by wage growth rates.

Table 7: Second-Stage: VA-based Shift-Share Instrument (11) Applied to Declining Wage vs. Growing Wage Regions

Dependent Variable:	(log) Elem.Ed.Exp.pp				
Model:	All	Hyper-Declining (Wage)	Declining (Wage)	Growing (Wage)	Hyper-Growing (Wage)
	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
(log) Weekly Wage	0.2088*** (0.0418)	0.3327*** (0.0599)	0.3579*** (0.0736)	0.1886*** (0.0456)	0.1369** (0.0692)
(11, log) Elem.Ed.Exp.pp	0.5219*** (0.0150)	0.5338*** (0.0238)	0.5684*** (0.0284)	0.5147*** (0.0168)	0.5412*** (0.0282)
(log) IG Revenue pp	0.2148*** (0.0199)	0.2627*** (0.0292)	0.1980*** (0.0269)	0.2170*** (0.0225)	0.1526*** (0.0364)
(log) Real GDP Priv. Industry pc	0.0589*** (0.0143)	-0.0042 (0.0225)	0.0013 (0.0298)	0.0645*** (0.0149)	0.0687*** (0.0152)
(log) Enrollment	-0.1729*** (0.0143)	-0.2020*** (0.0246)	-0.1819*** (0.0336)	-0.1707*** (0.0158)	-0.1767*** (0.0302)
% Black	0.0760 (0.1514)	-0.0452 (0.2242)	-0.1709 (0.2449)	0.1608 (0.1811)	0.5420 (0.4769)
% Hispanic	0.1265 (0.1198)	0.3644* (0.2032)	0.3595** (0.1618)	0.0906 (0.1410)	0.2160 (0.2408)
<i>Fixed-effects</i>					
unit	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	13,356	3,339	1,680	11,676	3,339
R ²	0.90746	0.92022	0.94124	0.89968	0.90294
Within R ²	0.53452	0.57958	0.60432	0.52453	0.52100
Wu-Hausman	48.367	15.082	13.615	36.536	18.645
Wu-Hausman, p-value	3.71×10^{-12}	0.00011	0.00023	1.55×10^{-9}	1.62×10^{-5}
Wald (IV only)	24.964	30.802	23.639	17.085	3.9135
Wald (IV only), p-value	5.91×10^{-7}	3.08×10^{-8}	1.27×10^{-6}	3.6×10^{-5}	0.04798
F-test (1st stage), (log) Weekly Wage	10,125.5	2,172.6	1,290.1	8,825.9	2,120.2
F-test (1st stage), p-value, (log) Weekly Wage	0×10^{-16}	0×10^{-16}	0×10^{-16}	0×10^{-16}	0×10^{-16}

Clustered (unit) standard-errors in parentheses
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 8: Second-Stage: VA-based Shift-Share Instrument (11) Applied to Declining GDP vs. Growing GDP Regions

Dependent Variable:	(log) Elem.Ed.Exp.pp				
Model:	All	Hyper-Declining (GDP)	Declining (GDP)	Growing (GDP)	Hyper-Growing (GDP)
	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
(log) Weekly Wage	0.2088*** (0.0418)	0.2890*** (0.0655)	0.2956*** (0.0568)	0.1689*** (0.0502)	0.2231*** (0.0735)
(11, log) Elem.Ed.Exp.pp	0.5219*** (0.0150)	0.5571*** (0.0202)	0.5416*** (0.0183)	0.5061*** (0.0209)	0.5000*** (0.0273)
(log) IG Revenue pp	0.2148*** (0.0199)	0.2258*** (0.0357)	0.2512*** (0.0305)	0.2001*** (0.0261)	0.1829*** (0.0369)
(log) Real GDP Priv. Industry pc	0.0589*** (0.0143)	0.0250 (0.0320)	0.0251 (0.0286)	0.0670*** (0.0161)	0.0774*** (0.0169)
(log) Enrollment	-0.1729*** (0.0143)	-0.1706*** (0.0282)	-0.1625*** (0.0200)	-0.1856*** (0.0194)	-0.2216*** (0.0282)
% Black	0.0760 (0.1514)	0.0422 (0.1993)	-0.0057 (0.1596)	0.2340 (0.2721)	0.0563 (0.6273)
% Hispanic	0.1265 (0.1198)	-0.0214 (0.1878)	0.0627 (0.1382)	0.1502 (0.1548)	0.1968 (0.1765)
<i>Fixed-effects</i>					
unit	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	13,356	3,339	5,544	7,812	3,339
R ²	0.90746	0.90502	0.90793	0.90584	0.86334
Within R ²	0.53452	0.56408	0.57108	0.49633	0.49480
Wu-Hausman	48.367	12.836	22.519	21.691	21.815
Wu-Hausman, p-value	3.71×10^{-12}	0.00035	2.14×10^{-6}	3.26×10^{-6}	3.13×10^{-6}
Wald (IV only)	24.964	19.485	27.096	11.318	9.2212
Wald (IV only), p-value	5.91×10^{-7}	1.05×10^{-5}	2.01×10^{-7}	0.00077	0.00241
F-test (1st stage), (log) Weekly Wage	10,125.5	3,009.0	4,379.7	5,685.7	2,520.5
F-test (1st stage), p-value, (log) Weekly Wage	0×10^{-16}	0×10^{-16}	0×10^{-16}	0×10^{-16}	0×10^{-16}

Clustered (unit) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

3.2.2 State-by-state estimation

Next, given the substantial heterogeneity in state-level economic makeup and public finance regimes, we investigate state-specific relationships between our variables of interest.

States vary in the number of commuting zones they contain. Figure 12 demonstrates that states have anywhere between 2 (Delaware) and 58 (Texas) commuting zones. This allows us to estimate panel-style regressions within each state to net out between-state variation that might be confounding our current treatment estimates. We exclude any states that contain less than 5 commuting zones (Delaware, Maine, Nevada, New Hampshire, New Jersey) due to sample size concerns.

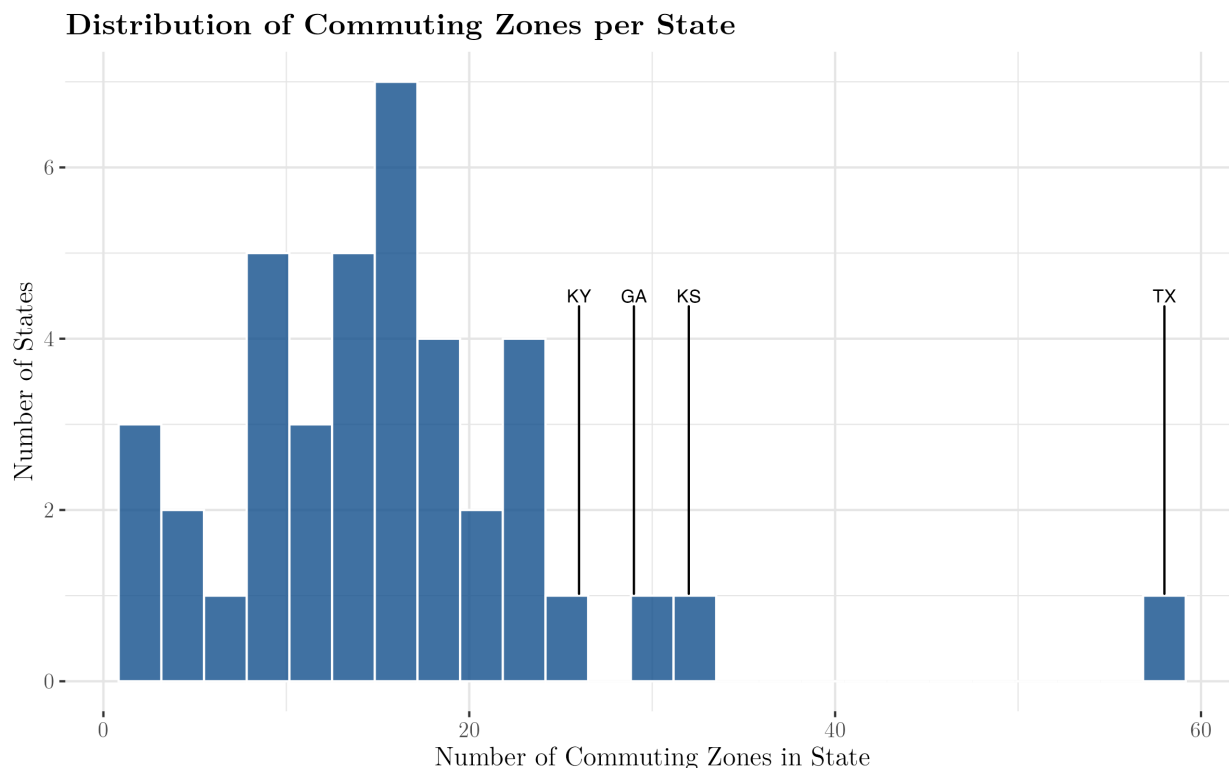
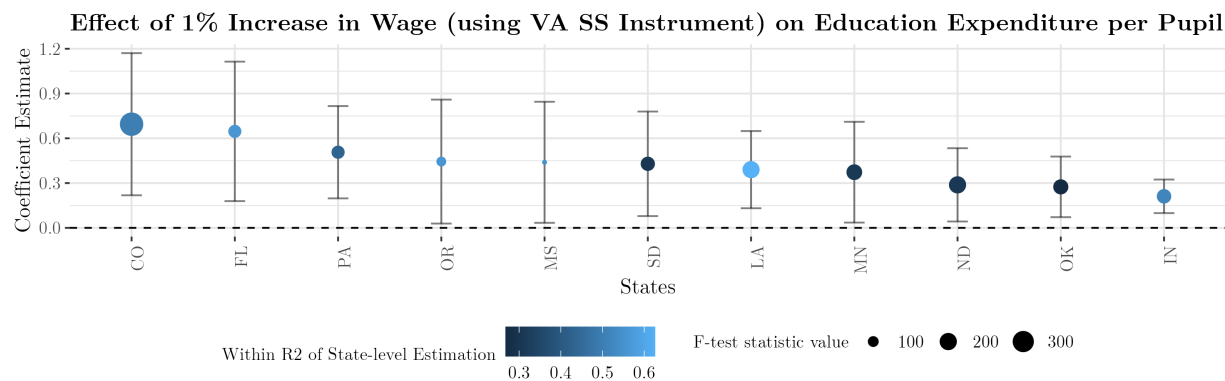


Figure 8: Histogram: Commuting Zones by State

Using our instrumental variable approach with a value-added based shift-share instrument, we corroborate the directionality and magnitude of the effect for 10 states: Colorado, Florida, South Dakota, Kentucky, North Dakota, Louisiana, Pennsylvania, Oregon, Oklahoma, and Indiana. In estimating the state-by-state regressions, we only present the results for states that exceed conventional weak instrument thresholds (F statistic ≥ 12 and p-value < 0.05) and whose p-values for the second stage coefficient of interest is < 0.1 .

Examining various characteristics of these states, we find that they vary widely in demographic composition, enrollment levels, and wage levels indicating that the detected effects are not attributable to any extremes in these values. Notably, these states even vary in reliance on local versus intergovernmental sources for public education expenditure.



Effect of 1% increase in Wage using 11 of VA SS Instrument on PP Education Expenditure controlling for enrollment, GDP, % black, % hispanic, and intergovernmental transfers. Within R² of state-level estimation reflected in color of point. Displays only states whose second-stage coefficient is statistically significant at the 10% level and first-stage F statistic ≥ 12 and p-value < 0.05

State Rankings Across Socioeconomic Indicators

All 35 states included in the state-by-state estimation are ranked by the population-weighted mean across commuting zones within the state. Each variable is ranked from highest (top) to lowest (bottom) value.

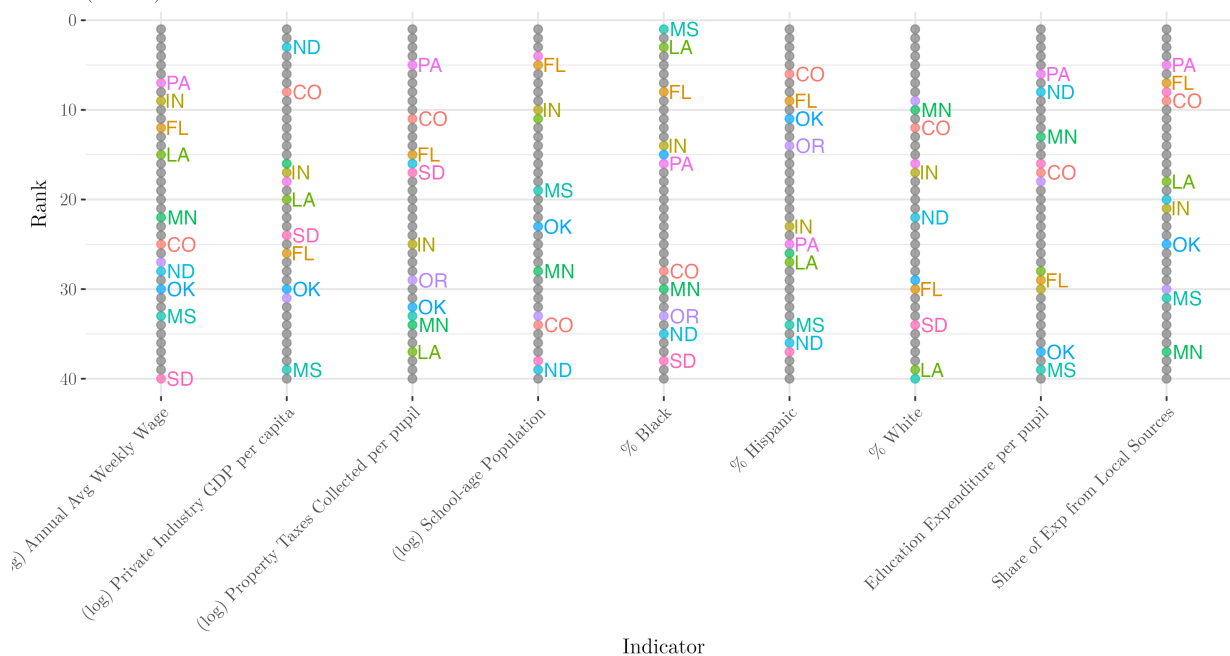


Figure 9: State-by-State Wage Effect Using VA SS Shock

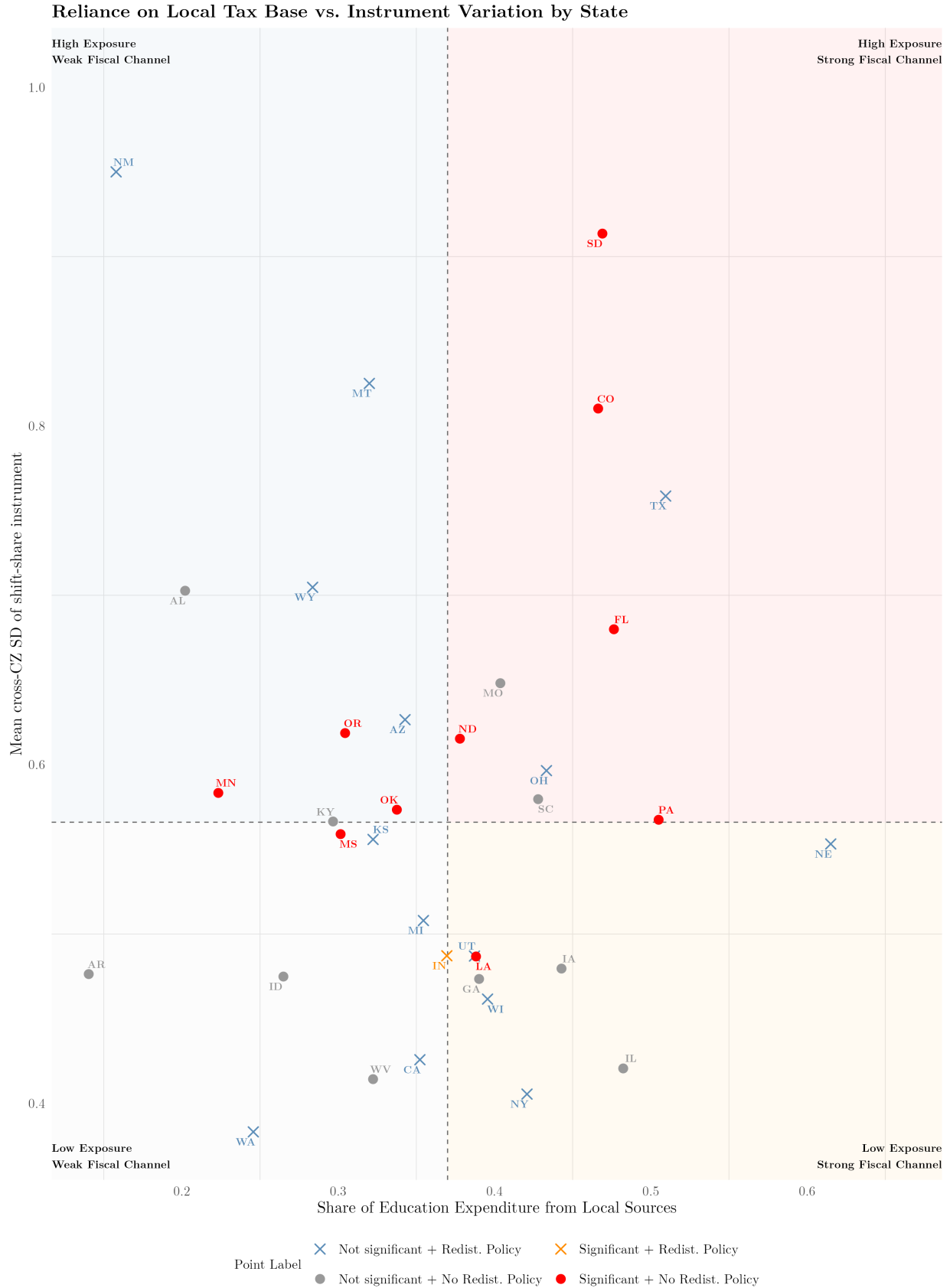


Figure 10: Wage Variation, Property Tax Reliance, and Redistributive Policies

4 Conclusion

Health care and social assistance sector is the dominant employer of most commuting zones in the country. Therefore, states susceptible to wage-related fiscal losses or gains are those that rely primarily or disproportionately on sectors other than Health care and social assistance. By design, the shift-share instrument detects sudden changes in industry-level wages. Industries characterized by such sudden shifts are the Construction, Mining, and Agricultural industries. Analyzing the states in which we detect a statistically significant causal response of public education budgets to wage changes, we indeed see that several commuting zones rely disproportionately on these more volatile sectors. Furthermore, the role of intergovernmental revenues is ambiguous across these states. In other words, though intergovernmental revenues have the potential to equalize funding across heterogeneous commuting zones within a state, depending on the symmetry of the mechanism's response to positive or negative economic shocks and its susceptibility to broader economic changes (regional or national-level) shocks, they can either insulate commuting zones from adverse economic forces, or not. This latter point is a particularly relevant distinction as the data emphasizes the critical effect that the 2008 financial crisis and subsequent recession had on local public education budgets. This effect was likely driven through a direct depression of property values due to the collapse of the mortgage market as well as a wage-based effect coming from widespread unemployment and the closure of local employment facilities. In other words, global economic depressions or shocks are likely to impact local school districts regardless of the structure of local funding (local versus intergovernmental revenues). In states in which local revenues make up a disproportionate share of public education spending, global economic conditions are likely to impact budgets heterogeneously depending on the degree of connectedness between local industrial composition and those industries primarily affected. On the other hand, in states where intergovernmental revenue play a relatively greater role in supporting public education spending, state-level revenues are likely to be affected by more general economic conditions, like a global financial crisis or depression. Altogether, this indicates that though intergovernmental revenues can be relied on to equalize both positive and negative impacts of economic transformation on local areas, the stability of these intergovernmental revenues (which are of course dependent on local economic conditions already) implicate the stability of local budgets in turn.

An interesting case example is Idaho where the centralization of public education funding through a greater reliance on intergovernmental funding did not alleviate spending shortfalls. In 2006, the Idaho state legislature passed House Bill 1 which reduced property tax burdens by \$260 million. State funding buffered this reduction for a period of two years. However, the advent of the 2008 Recession promptly depressed state funding leading to losses in local public education expenditure, similar to the fate of most states. Interestingly, this case demonstrates that the centralization of funding, though it might shield against unequal funding across communities within a state, broader economic conditions dictate public education funding in Idaho and other states. This reality explains why we do not attribute a causal local wage-spending relationship in Idaho, precisely because any local public education expenses trace the economic health of the state itself.

- Detroit, Michigan
- In certain states we can identify the effect of monopsonistic employers in specific areas (Rochester, MN IBM Campus; Marshall, MN sale of major local food employer Schwan's Company).
- Often related to construction booms - Brainerd, MN. Roseau, MN Polaris recreational sports equipment.
- These are places where wage data are dominated by a narrow set of employers or industries, where there is limited ability to absorb shocks, and where the exit or contraction of even one significant employer can push aggregate wages visibly downward.

Essentially, the paper presents a distillation of substantial state and sub-state level heterogeneity in industrial composition, relative vulnerability, and legacies of campaigns for greater equalization. Testing the strength of the constructed instrument by state reveals that structural wage losses impact public education budgets as long as the fiscal channel from property taxes is not broken. Furthermore, the 2008 Recession sits in the middle of the panel window, calling attention to the fact that even states with strong redistributive systems faced substantial public education losses as a result of a broader depressed economic state.

States in which we expect to detect a strong fiscal response to wage fluctuations (i.e., Texas, West Virginia, Montana), local safety nets via strong distribution mechanisms successfully counteract these forces. Whereas states with similarly exposed wage portfolios see co-movement between wages and fiscal stability due either to outsized instrument strength (in other words, extreme wage shocks) or the absence of adequate redistributive mechanisms.

This work intentionally inhabits a narrow scope. The causal claim derives from a narrow consideration of wage forces, one in which only those induced by demand shifts in a select few industries. The estimation strategy relies on local co-movement with global forces. However, local public purses are susceptible to myriad sources of fiscal uncertainty stemming not only from other types of wage fluctuations (changes in minimum wage laws, local firm and plant closures) but regional and national economic fluctuations, as evidenced by the dramatic impact of the 2008 Recession on all education revenue and spending variables in our sample. Essentially, whether the tax burden shifts from local to state sources as a method of equalization, state-level reliance on taxation of other indicators of economic health (sales, consumption, income) cannot, by design, fully insulate local areas from under-investment in local public education, though such under-investment would likely be more equitably spread than a system in which only local sources make up public education revenue sources.

Precisely because the national landscape is so diverse and contains 50 distinct public education systems that confound a reasonable average treatment effect, the analysis' conclusion requires careful consideration of the estimation strategy. We can conclude that local public education budgets are not insulated from global economic forces impacting local wages. This work demonstrates a statistically significant causal positive relationship between wages and public education spending both at a national level and in a few specific states. The focalization of the effect on ten specific states provide valuable insight into the variation driving the estimation strategy, namely, wage changes stemming from fluctuations in national-level demand for specific industries. Therefore, distinguishing between the states in which we have a strong instrument but weak or no second-stage effect on education spending allows for vantage points to uncover the policy landscapes that insulate “exposed” states from these fluctuations.

4.1 Discussion

This paper set out to examine whether local wage growth translates into higher local public education expenditure across US commuting zones, and whether this relationship varies across institutional environments, labor-market conditions, and long-run regional trajectories. Motivated by the persistent divergence between productivity growth and wage gains for typical workers, we asked whether the decoupling of wages from productivity has implications beyond household well-being and specifically, whether uneven wage growth shapes the fiscal capacity of communities to invest in public education.

We establish a causal link between local wage growth and public education expenditure. We show that public education expenditure is not insulated from local labor-market conditions. In the baseline specification, a 10 percent increase in local wages generates a 2.2 percent increase in education spending in the short run and a 4.6 percent increase in the long run. The dynamic components of the econometric specification indicate significant persistence in local public education budgets. This persistence likely reflects institutional inertia but also the infrequency with which local tax rates are revised, allowing local wage growth to boost education spending. It also suggests asymmetric adjustment: communities experiencing wage growth are likely to see gradual increases in education spending as higher incomes strengthen their revenue base, while declining communities could experience the opposite effect.

Importantly, the positive elasticity we document is not uniform across the country. It is concentrated in less than a third of the states in the sample (Colorado, Florida, South Dakota, Kentucky, Louisiana, Pennsylvania, North Dakota, Oregon, Oklahoma, and Indiana). Many other states exhibit minimal or insignificant responsiveness, suggesting that these risks are not uniformly distributed. These findings underscore that equalization programmes need to consider different types of risks facing communities when designing their work.

Though outcomes measured in this study are not direct inequality metrics, the results have clear implications for spatial disparities in local public well-being. Our findings reveal that the decentralized school financing

system in the US has the potential to exacerbate inequalities in local public well-being by failing to equalize across regions experiencing diverging growth trajectories. As a result, in the states in which local spending is responsive to changes in local wages, the quality of early childhood education might be compromised by macroeconomic trends and industry-specific shocks beyond local control. Thus, equalization formulas at the state and federal level that fail to account for wage trends and fiscal multipliers may contribute to disparities in public goods delivery. Theories of effective equalization, and indeed preferences for redistribution, differ especially across regions of the the United States. However, equalization efforts should at least aim to insulate communities from potential downward pressures on public education expenditure.

The determinants of inequality in public education delivery in the US are multiple and complex. Significant evidence exists of the role of historically discriminatory policies related to congressional districting in driving under-investment in low-income areas of color [trounstine2016, trounstine2021, sosina2019]. Though this work does not directly inform this debate, further work could explore the extent to which wage growth interacts with such structural policies.

Addressing structural inequality requires rethinking education finance, taking the decentralized American context as given. Ensuring educational equity for all children requires not just strengthening existing mechanisms for redistribution, but also, in light of our results, insulating communities from macroeconomic trends that impact communities heterogeneously.

4.2 Limitations

- Though we have attempted to bypass deep endogeneity issues in the relationship between wages and public education expenditure, our excludability assumptions about local industrial composition are strong.
- Commuting zones mask substantial within-commuting zone heterogeneity. Further work could apply the same estimation strategy to county-level observations to address this challenge.

5 Inventory of Remaining Work

Below, we outline the remaining work planned prior to pursuing journal submission. In addition to cosmetic improvements to the manuscript, we intend to:

- Provide an improved calculation about the economic meaning of the results. Though we provide a preliminary back-of-the-envelope calculation about the impact of a 2% change in expenditure per pupil in terms of staff costs, this calculation is likely simplified. We plan to make a comparison of the scale of potential changes to public education budgets in terms of staff, food, or other resource costs using data from the SLGF. Additionally, we might be able to speak to the potential for such a change in expenditure per pupil to reduce class sizes or fund salary increases for teachers, for example.
- Report the marginal F-statistic of the instrumental variable specifications to demonstrate the relative value of the instruments independent of the effect of the auto-regressive term in the first-stage. The autoregressive terms are likely inflating the F-statistic which is an important limitation to highlight in the main text.
- Report over-identification tests in regression tables.
- Include additional appendices reporting the results when instrumenting local property prices rather than local wage levels.
- Provide a robustness check using a cross-sectional empirical strategy to determine the relative importance of cross-sectional versus time variation in the sample.

6 Data and Code Availability

Code and data to reproduce the analysis will be made available on Zenodo. A working version of the database and code is currently public on [Github](#).

7 Acknowledgements

I would primarily like to thank Jennifer Castle for valuable supervision and comments throughout the course of this project. Thanks additionally to Jonas Kurlle and Kasper Pedersen of the Calleva Research Centre and Stefania Innocenti for valuable comments.

8 Use of AI

Below, we provide a summary of the tasks where AI was used. We limited its use to debugging and efficiency improvements during data cleaning and analysis, formatting of tables and figures, and helping with compatibility between various typesetting tools. We did *not* use AI to generate any text for this manuscript.

- Used ChatGPT or Claude to help improve readability of plots (formatting, margins, labeling, font size, distinguishing between model fit and testing values).
- Used ChatGPT and Claude to debug errors in R during data cleaning and plotting.
- Used ChatGPT to improve compatibility between Quarto Markdown and Latex integrations.
- Used ChatGPT to provide suggestions for reducing run time of repetitive tasks (ex. downloading and processing multiple data files).